

Chapter 4 Trigonometry:

4.1 Radian and Degree Measure:

* **Trigonometry**:- The word trigonometry derived from the Greek language which means 'measurement of triangles'.

It is branch of Mathematics concerned with relationship b/w the sides and angles of triangles.

* **Angle**:- An angle is formed when two rays or lines meet at common endpoint.

The common point is called the vertex of an angle.

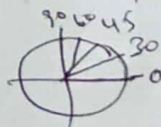
Types of Angle:-

- 1) **Acute Angle**: It lies b/w 0° to 90° .
- 2) **Obtuse Angle**: It lies b/w 90° to 180° .
- 3) **Right Angle**:- The angle which is exactly equal to 90° .
- 4) **Straight Angle**:- The angle which is exactly equal to 180° .
- 5) **Reflex angle**:- The angle which is greater than 180° and less than 360° .
- 6) **Complete angle**:- The complete rotation of angle equal to 360° .
- 7) **Positive angle**:- An angle measured in Anti clockwise direction.
- 8) **Negative Angle**:- An angle measured in clockwise direction.
- 9) **Coterminal angles**:- Two Angles have the same initial and terminal sides, such angles are coterminal angles.

* **Measure of an angle**:- The amount of rotation from the initial side to the terminal side determines the measure of an angle.

⇒ **Degree of an angle**:- It is represented by $^\circ$ (read as degree).

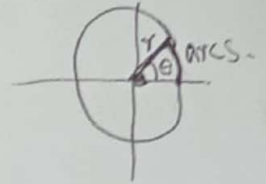
An angle is said to be equal to 1° , if rotation from the initial to the terminal side is equal to $\frac{1}{360}$ of the full rotation.



* Radian Measure :-

The length of arc 's' of a unit circle is numerically equal to the measurement in radian of the angle θ .

$$\theta = \frac{s}{r} \quad \text{where } \theta \text{ is measured in radian.}$$



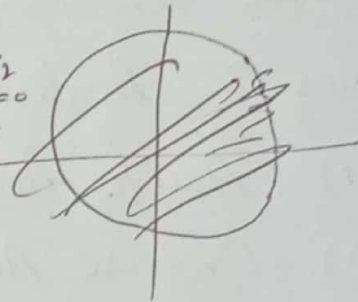
In a complete circle, there are 2π radians.

$$360^\circ = 2\pi \text{ rad}$$

$$1 \text{ rad} = \frac{360^\circ}{2\pi}$$

$$1 \text{ rad} = \frac{180^\circ}{\pi}$$

$$\begin{array}{l} \theta = \frac{\pi}{2} \quad \text{1st} \\ \frac{\pi}{2} < \theta < \pi \\ \theta = \pi \\ \pi < \theta < \frac{3\pi}{2} \\ \theta = \frac{3\pi}{2} \end{array}$$

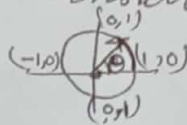


Exp 01, 2, 3, 4

* 4.2 Trigonometric functions: The Unit circle

→ The Unit circle: Consider the unit circle given by

$$x^2 + y^2 = 1$$



(1st do this)

* Trigonometric Functions: Let t be a real number and $P(x, y)$ be the point on the unit circle corresponding to t .

$$\sin t = y$$

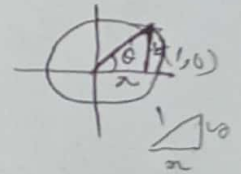
$$\cos t = x$$

$$\tan t = \frac{y}{x} \quad x \neq 0$$

$$\csc t = \frac{1}{y}$$

$$\sec t = \frac{1}{x}$$

$$\cot t = \frac{x}{y}$$

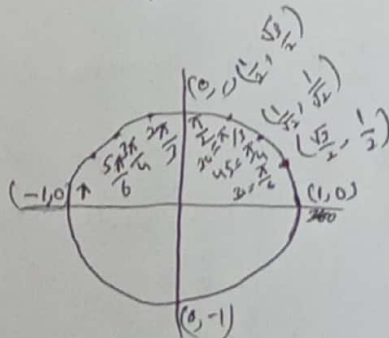
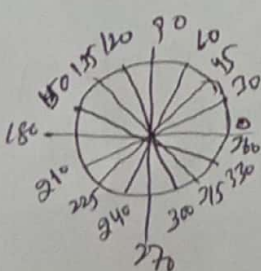


$$\sin \theta = \frac{P}{H}$$

$$\cos \theta = \frac{B}{H}, \quad \tan \theta = \frac{P}{B}$$

$$\cot \theta = \frac{B}{P}, \quad \csc \theta = \frac{H}{P}$$

$$\sec \theta = \frac{H}{B}$$



Exp :- 01,